

Eva Dimitrova

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EDUCATION

University of Maryland

Bachelor of Science, Computer Engineering: 3.6 GPA

College Park, MD

Expected May 2028

Languages: Java, Python, JavaScript, C/C++ | **Backend & APIs:** FastAPI, Flask, Node.js, REST APIs

Databases & Data: SQL, Supabase (Postgres), FAISS (vector DB) | **Frontend:** React | **Systems & Tools:** Git, Linux

TECHNICAL EXPERIENCE

Software Engineering Intern (Incoming)

Liberty Mutual Insurance

Portsmouth, NH

May - August 2026

Research Intern | Python, Excel

Maryland Information and Network Dynamics Lab

College Park, MD

May - August 2025

- Developed **anomaly detection models** for building energy data, reducing false positives by **60%**.
- Automated visualization pipelines generating daily reports for **30+ meters**, cutting manual analysis time from **hours to seconds**.
- Standardized raw sensor feeds into **query-ready datasets**, enabling fast analytical queries across time-series energy data for 15+ researchers.

Computer Engineering Intern | C, C++, Linux, seL4

Brown Deer Technology

Bel Air, MD

May - August 2025

- Optimized a **C based libconfig parser** on Linux, improving automation and testing efficiency by **50%**.
- Investigated and validated the **seL4 microkernel on RISC-V**, debugging low level system behaviors to strengthen deployment and testing processes.
- Enhanced expertise in **embedded systems, Linux internals, and kernel-level debugging** through performance tracing and real time analysis.

ECE Department Peer Mentor

A. James Clark School of Engineering

College Park, MD

September 2025 - Present

- Mentored 50+ undergraduate students** in engineering coursework, debugging strategies, and project planning.

PROJECTS

YouTube AI Video Organizer | Python, FastAPI, Sentence Transformers, FAISS, JavaScript

January 2026

- Designed **data ingestion, indexing, and storage workflows** to support scalable semantic search over large, unstructured datasets.
- Deployed an **AI powered search system** that semantically indexes and retrieves YouTube liked videos using vector embeddings and cosine similarity.
- Designed a **FastAPI backend with RESTful endpoints** for scalable search and metrics reporting.
- Automated **YouTube Data API ingestion pipelines**, eliminating manual categorization and **reducing discovery time by 70%**.

Student Performance Predictor Model | Python, React, Flask, PyTorch, XGBoost

August 2025

- Built an end-to-end ML pipeline including data preprocessing, feature engineering, model training, and deployment.
- Preprocessed 30+ columns** by cleaning missing values, mapping target labels, removing duplicates, and encoding categorical variables using one-hot encoding.
- Developed an XGBoost-based model to predict student performance, **achieving 91% accuracy on held-out validation data**.
- Created a dynamic user interface using **React + Flask** to effectively showcase results.

Study Group Finder Web App | React, Supabase, SQL

January 2025

- Built a responsive, multi-page React app where users can sign up, log in, create study groups, and join existing groups.
- Integrated **Supabase** backend services and wrote **SQL queries** to support core CRUD operations for users and group membership data.
- Implemented **state driven UI updates** tied to backend data changes, enabling real time reflection of authentication state and group actions.

ACTIVITIES

Capital One Fintech Future Hackathon | Node.js, React, C++, OAuth, SQL

May 2025

Full-Stack Engineer

- Collaborated with a team of 5 to build *Savings Buddy*, a React web app for budgeting with goal streaks and peer competition.
- Built front-end features and integrated **Plaid API via Node.js backend** to enable real-time goal tracking, delivering a **fully functional MVP in one week**.

App Development Club | Python, AWS Bedrock

February - May 2025

Machine Learning Engineer

- Worked alongside a team to develop a **RAG pipeline** using LLMs for financial data retrieval, automating structured data extraction and improving team access to complex datasets.
- Contributed to research on **tabular data extraction** methods and participated in testing **AWS Bedrock** workflows to **support ingestion of structured financial formats**.